10. $P_A^o = 10$ torr , $P_{solution} = 9$ torr

$W_A = 20$ g $\quad W_B = 1$ g

$M_A = 200$ g mol^{-1} $M_B = ?$

$$\frac{\Delta P}{P_A^o} = \frac{W_B \times M_A}{M_B \times W_A}$$

$$\frac{10-9}{10} = \frac{1 \times 200}{M_B \times 20}$$

$$M_B = \frac{200}{20} \times 10 = 100 \text{ g mol}^{-1}$$

11. $W_2 = 2.56$ g

$W_1 = 100$ g

$T = 319.692$ K

$K_b = 2.42$ K Kg mol^{-1}

$\Delta T_b = (319.692 - 319.450)$ K

$= 0.242$ K

$$M_2 = \frac{K_b \times W_2 \times 100}{\Delta T_b \times W_1}$$

$$= \frac{2.42 \times 2.56 \times 1000}{0.242 \times 100}$$

$$M_2 = 256 \text{ g mol}^{-1}$$

Molecular mass of sulphur in solution
$= 256$ g mol^{-1}

atomic mass of one mole of sulphur atom = 32

No. of atoms in a molecule of sulphur

$$= \frac{256}{32} = 8$$

Hence molecular formula of sulphur is S_8.

12. $W_2 = 2$g $\quad W_1 = 75$ g

$\Delta T_f = 0.2$ K $\quad K_f = 5.12$ K Kg mol^{-1}

$M_2 = ?$

$$M_2 = \frac{K_f \times W_2 \times 1000}{\Delta T_f \times W_1} = \frac{5.12 \times 2 \times 1000}{0.2 \times 75}$$

$$= 682.66 \text{ g mol}^{-1}$$

13. Osmotic pressure of urea solution (π_1)=CRT

$$= \frac{W_2}{M_2 V} RT$$

$$= \frac{6}{60 \times 1} \times RT$$

Osmotic pressure of glucose solution

$$(\pi_2) = \frac{W_2}{180 \times 1} \times RT$$

For isotonic solution,

$\pi_1 = \pi_2$

$$\frac{6}{60} RT = \frac{W_2}{180} RT$$

$$\Rightarrow W_2 = \frac{6}{60} \times 180$$

$W_2 = 18$ g

14. $$i = \frac{\text{Observed property}}{\text{Theoritical property (calculated)}}$$

Given $\Delta T_f = 0.680$ K

$m = 0.2$ m

ΔT_f (observed) $= 0.680$ K

ΔT_f (calculated) $= K_f \cdot m$

$= 1.86$ K Kg mol^{-1} $\times$ 0.2 mol Kg^{-1}

$= 0.372$ K

$$i = \frac{(\Delta T_f) \text{ Observed}}{(\Delta T_f) \text{ Calculated}} = \frac{0.680 \text{ K}}{0.372 \text{ K}} = 1.82$$

MCQ

1. molality $= \dfrac{\text{number of moles of solute}}{\text{weight of solvent (in kg)}}$

$$= \frac{\left(\frac{1.8}{180}\right)}{0.25} = \frac{0.01}{0.25} = 0.04 \text{ M}$$

option (d) is correct.

2. option (d) is correct. Molality and mole fraction are independent of temperature.

3. $M_1 \times V_1 = M_2 \times V_2$ [$\because$ 0.1 M $Al(OH)_3$ gives $3 \times 0.1 = 0.3$ M OH^- ions]

$0.3 \times V_1 = 0.1 \times 21$

$$V_1 = \frac{0.1 \times 21}{0.3} = 7 \text{ ml}$$

option (b)

Key answers -jagan.indd 299 22-01-2021 13:23:28